

C:\Users\91769\OneDrive\Desktop\study_files\OS\C lab\EXP-1.cpp - [Executing] - Dev-C++ 5.11

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TDM-GCC 4.9.2 64-bit Release

(globals)

Project Classes Debug

EXP-7.cpp EXP-8.cpp EXP-9.cpp EXP-10.cpp EXP-11.cpp

EXP-13.cpp EXP-12.cpp EXP-14.cpp EXP-15.cpp EXP-16.cpp EXP-17.cpp EXP-18.cpp EXP-19.cpp EXP-20.cpp EXP-1.cpp EXP-2.cpp EXP-3.cpp EXP-4.cpp EXP-5.cpp EXP-6.cpp

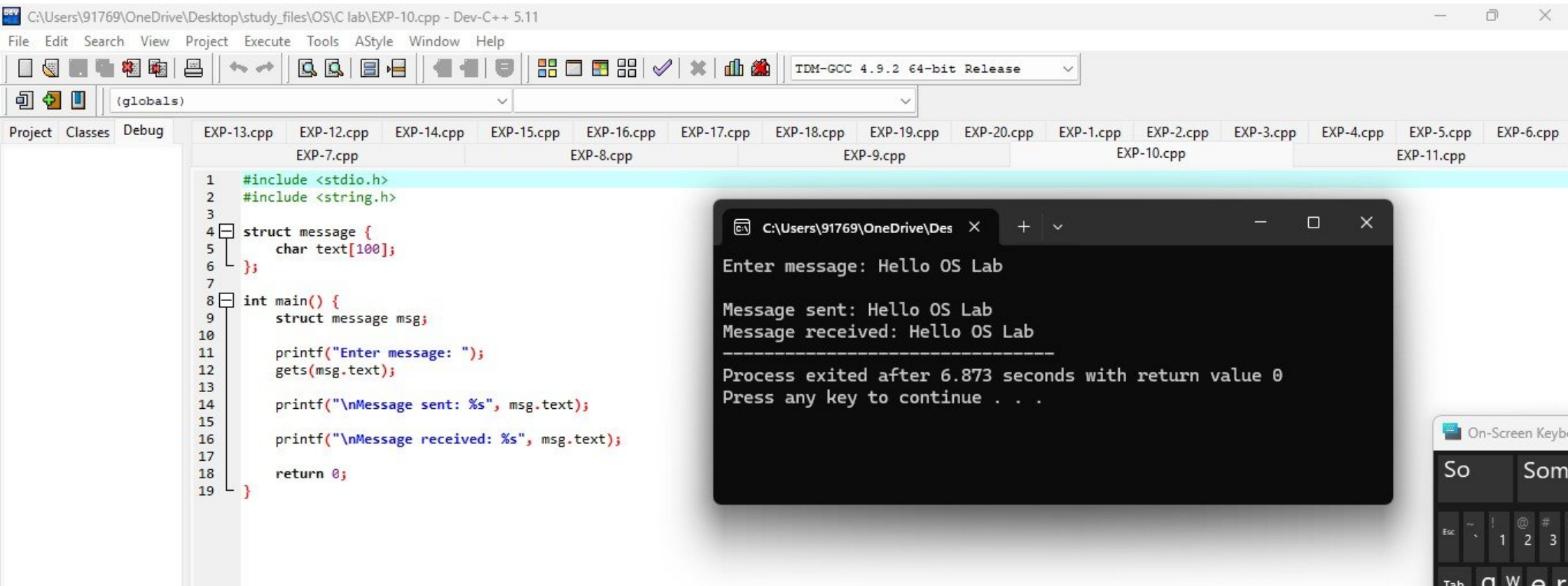
```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 int main() {
5     int parent_pid = 1001;
6     int child_pid = 1002;
7
8     printf("Simulating Process Creation\n");
9
10    printf("\nParent Process:\n");
11    printf("Parent PID: %d\n", parent_pid);
12    printf("Child PID: %d\n", child_pid);
13
14    printf("\nChild Process:\n");
15    printf("Child PID: %d\n", child_pid);
16    printf("Parent PID: %d\n", parent_pid);
17
18    return 0;
19 }
```

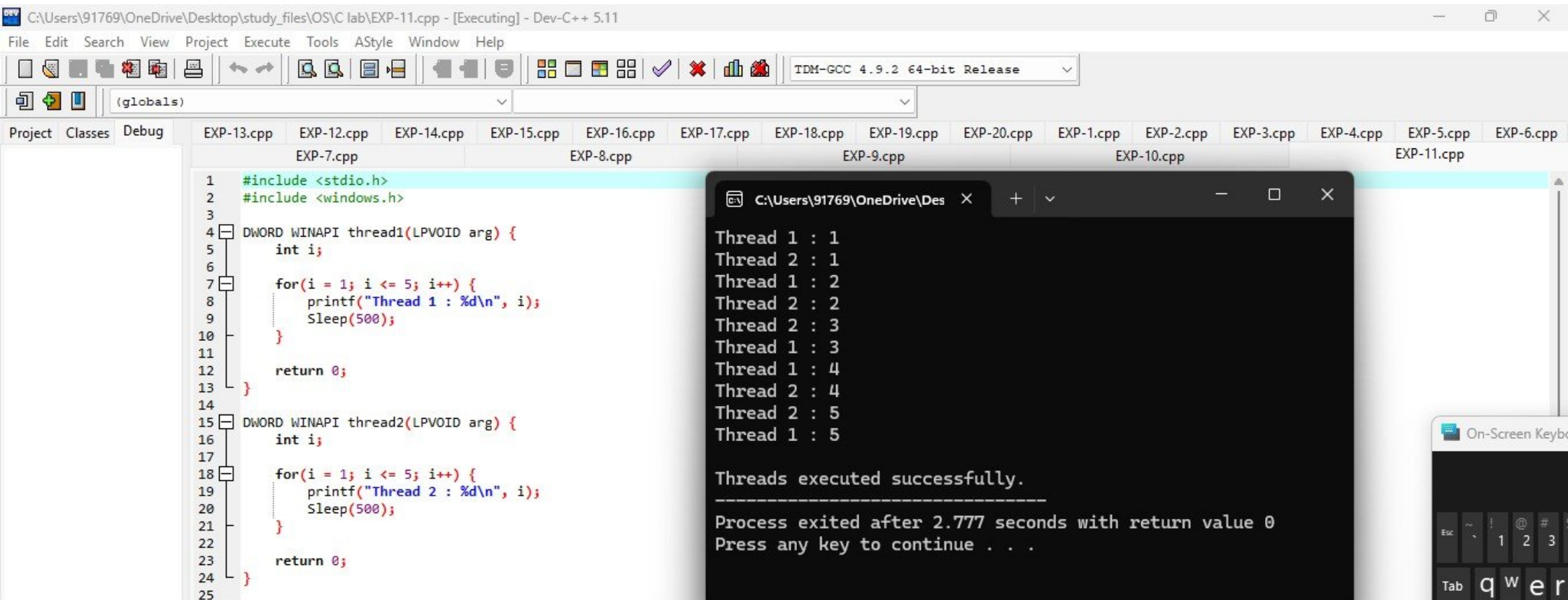
C:\Users\91769\OneDrive\Desktop

Parent Process:
Parent PID: 1001
Child PID: 1002

Child Process:
Child PID: 1002
Parent PID: 1001

Process exited after 0.1989 seconds with return value 0
Press any key to continue . . .





```
C:\Users\91769\OneDrive\Desktop\study_files\OS\C lab\EXP-12.cpp - [Executing] - Dev-C++ 5.11
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(globals)
Project Classes Debug
EXP-7.cpp EXP-8.cpp EXP-9.cpp
EXP-13.cpp EXP-12.cpp EXP-14.cpp EXP-15.cpp EXP-16.cpp EXP-17.cpp EXP-18.cpp EXP-19.cpp
40 for(i = 0; i < 5; i++) {
41     chopstick[i] = CreateMutex(NULL, FALSE, NULL);
42 }
43
44 // Create philosopher threads
45 for(i = 0; i < 5; i++) {
46     id[i] = i + 1;
47
48     ph[i] = CreateThread(
49         NULL,
50         0,
51         philosopher,
52         &id[i],
53         0,
54         NULL
55     );
56 }
57
58
59 // Wait for all philosophers to finish
60 WaitForMultipleObjects(5, ph, TRUE, INFINITE);
61 printf("\n\nAll philosophers finished dining.\n");
62
63
```

```
C:\Users\91769\OneDrive\Des
Philosopher 1 is THINKING
Philosopher 4 is THINKING
Philosopher 3 is THINKING
Philosopher 5 is THINKING
Philosopher 2 is THINKING
Philosopher 2 is HUNGRY
Philosopher 2 is EATING
Philosopher 3 is HUNGRY
Philosopher 1 is HUNGRY
Philosopher 5 is HUNGRY
Philosopher 4 is HUNGRY
Philosopher 1 is EATING
Philosopher 2 finished EATING
Philosopher 5 is EATING
Philosopher 1 finished EATING
Philosopher 5 finished EATING
Philosopher 4 is EATING
Philosopher 4 finished EATING
Philosopher 3 is EATING
Philosopher 3 finished EATING

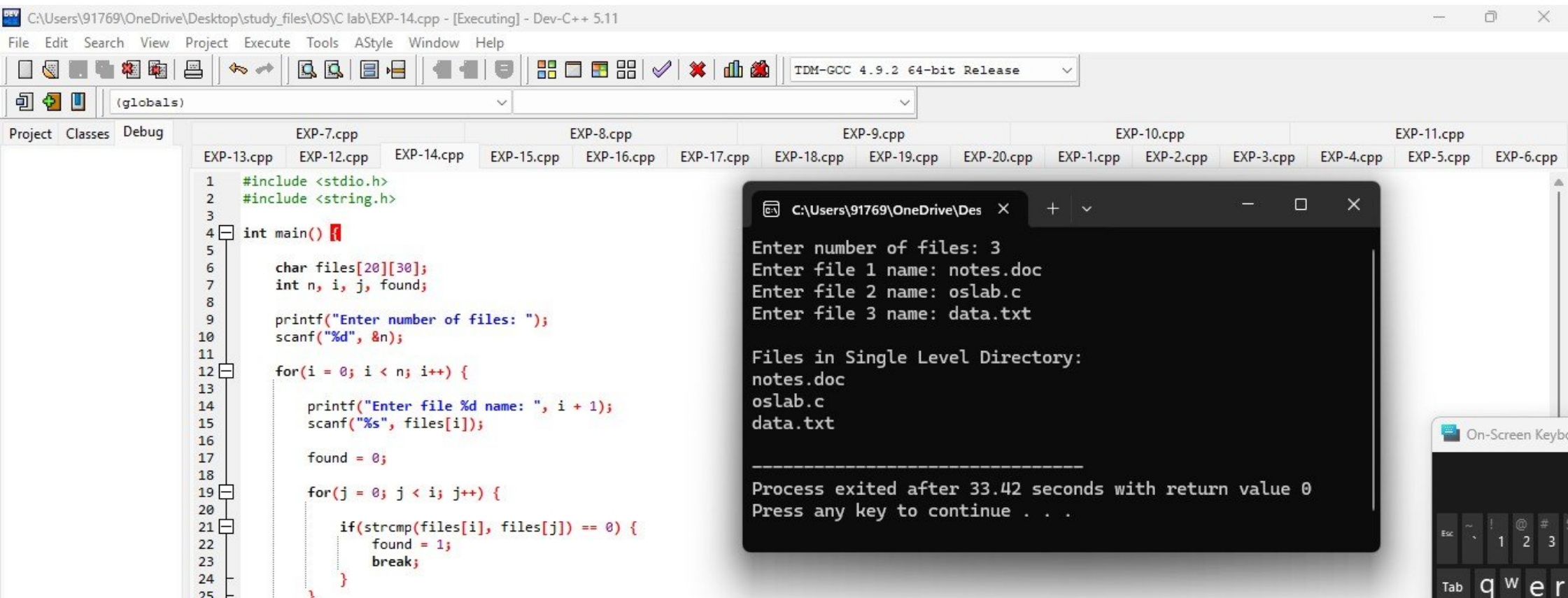
All philosophers finished dining.
```



```
C:\Users\91769\OneDrive\Desktop\study_files\OS\C lab\EXP-13.cpp - [Executing] - Dev-C++ 5.11
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TDM-GCC 4.9.2 64-bit Release
(globals)
Project Classes Debug
EXP-7.cpp EXP-8.cpp EXP-9.cpp
EXP-13.cpp EXP-12.cpp EXP-14.cpp EXP-15.cpp EXP-16.cpp EXP-17.cpp EXP-18.cpp EXP-19.cpp
1 #include <stdio.h>
2
3 int main() {
4     int mem[20], process[20];
5     int i, j, n, m;
6     int allocated[20] = {0};
7
8     printf("Enter number of memory blocks: ");
9     scanf("%d", &n);
10
11     for(i = 0; i < n; i++) {
12         printf("Enter size of block %d: ", i + 1);
13         scanf("%d", &mem[i]);
14     }
15
16     printf("Enter number of processes: ");
17     scanf("%d", &m);
18
19     for(i = 0; i < m; i++) {
20         printf("Enter size of process %d: ", i + 1);
21         scanf("%d", &process[i]);
22     }
23
24     printf("\n--- First Fit Allocation ---\n");
```

```
Enter number of memory blocks: 5
Enter size of block 1: 100
Enter size of block 2: 500
Enter size of block 3: 200
Enter size of block 4: 300
Enter size of block 5: 600
Enter number of processes: 4
Enter size of process 1: 212
Enter size of process 2: 417
Enter size of process 3: 112
Enter size of process 4: 426
```

```
--- First Fit Allocation ---
Process 1 allocated to Block 2
Process 2 allocated to Block 5
Process 3 allocated to Block 2
Process 4 cannot be allocated
```



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(globals)

Project Classes Debug

EXP-7.cpp

EXP-8.cpp

EXP-9.cpp

EXP-13.cpp

EXP-12.cpp

EXP-14.cpp

EXP-15.cpp

EXP-16.cpp

EXP-17.cpp

EXP-18.cpp

EXP-19.cpp

EXP-20.cpp

```
16     printf("\nEnter name of user %d: ", i + 1);
17     scanf("%s", user[i]);
18
19     printf("Enter number of files for %s: ", user[i]);
20     scanf("%d", &files);
21
22     for(j = 0; j < files; j++) {
23
24         printf("Enter file %d name: ", j + 1);
25         scanf("%s", file[i][j]);
26     }
27
28
29     printf("\n--- Two Level Directory Structure ---\n");
30
31     for(i = 0; i < users; i++) {
32
33         printf("\nUser: %s\n", user[i]);
34
35         for(j = 0; file[i][j][0] != '\0'; j++) {
36             printf("%s\n", file[i][j]);
37         }
38     }
39
40     return 0;
```

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Enter number of users: 2

Enter name of user 1: ram
Enter number of files for ram: 2
Enter file 1 name: data.txt
Enter file 2 name: lab.c

Enter name of user 2: hemanth
Enter number of files for hemanth: 2
Enter file 1 name: capstone.pptx
Enter file 2 name: report.doc

--- Two Level Directory Structure ---

User: ram
data.txt
lab.c

User: hemanth
capstone.pptx
report.doc

On-Screen Keybo

Esc ~ ! @ # \$ % ^ & * () _ + = , . / : ; " ' [\] { } | ~ ` 1 2 3 4 5 6 7 8 9 0 ~ ` q w e r

C:\Users\91769\OneDrive\Desktop\study_files\OS\C lab\EXP-16.cpp - [Executing] - Dev-C++ 5.11

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TDM-GCC 4.9.2 64-bit Release

(globals)

Project Classes Debug

EXP-7.cpp EXP-8.cpp EXP-9.cpp
EXP-13.cpp EXP-12.cpp EXP-14.cpp EXP-15.cpp EXP-16.cpp EXP-17.cpp EXP-18.cpp

```
16  
17     printf("Enter number of employees: ");  
18     scanf("%d", &n);  
19  
20     for(i = 0; i < n; i++) {  
21  
22         printf("\nEnter Employee ID: ");  
23         scanf("%d", &e.id);  
24  
25         printf("Enter Employee Name: ");  
26         scanf("%s", e.name);  
27  
28         printf("Enter Employee Salary: ");  
29         scanf("%f", &e.salary);  
30  
31         fwrite(&e, sizeof(e), 1, fp);  
32     }  
33  
34     fclose(fp);  
35  
36     fp = fopen("employee.dat", "rb");  
37  
38     printf("\nEmployee Records:\n");  
39
```

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```
Enter number of employees: 2  
  
Enter Employee ID: 101  
Enter Employee Name: hemanth  
Enter Employee Salary: 80000  
  
Enter Employee ID: 102  
Enter Employee Name: deebpesh  
Enter Employee Salary: 75000  
  
Employee Records:  
  
ID: 101  
Name: hemanth  
Salary: 80000.00  
  
ID: 102  
Name: deebpesh  
Salary: 75000.00
```


C:\Users\91769\OneDrive\Desktop\study_files\OS\C lab\EXP-2.cpp - Dev-C++ 5.11

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TDM-GCC 4.9.2 64-bit Release

(globals)

Project Classes Debug

EXP-7.cpp EXP-8.cpp EXP-9.cpp EXP-10.cpp EXP-11.cpp
EXP-13.cpp EXP-12.cpp EXP-14.cpp EXP-15.cpp EXP-16.cpp EXP-17.cpp EXP-18.cpp EXP-19.cpp EXP-20.cpp EXP-1.cpp EXP-2.cpp EXP-3.cpp EXP-4.cpp EXP-5.cpp EXP-6.cpp

```
1  #include <stdio.h>
2
3  int main() {
4      FILE *source, *destination;
5      char ch;
6
7      // Open source file
8      source = fopen("source.txt", "r");
9
10     if (source == NULL) {
11         printf("Source file not found.\n");
12         return 1;
13     }
14
15     // Open destination file
16     destination = fopen("destination.txt", "w");
17
18     // Copy contents
19     while ((ch = fgetc(source)) != EOF) {
20         fputc(ch, destination);
21     }
22
23     printf("File copied successfully.\n");
24
25     // Close files
```

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File copied successfully.

Process exited after 0.3033 seconds with return value 0
Press any key to continue . . .

C:\Users\91769\OneDrive\Desktop\study_files\OS\C lab\EXP-3.cpp - [Executing] - Dev-C++ 5.11

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(globals)

Project Classes Debug

EXP-7.cpp EXP-8.cpp
EXP-13.cpp EXP-12.cpp EXP-14.cpp EXP-15.cpp EXP-16.cpp EXP-17.cpp

```
1  #include <stdio.h>
2
3  int main() {
4      int n, i;
5      int bt[20], wt[20], tat[20];
6      float avg_wt = 0, avg_tat = 0;
7
8      printf("Enter number of processes: ");
9      scanf("%d", &n);
10
11     for(i = 0; i < n; i++) {
12         printf("Enter Burst Time for Process %d: ", i + 1);
13         scanf("%d", &bt[i]);
14     }
15
16     wt[0] = 0;
17
18     for(i = 1; i < n; i++) {
19         wt[i] = wt[i - 1] + bt[i - 1];
20     }
21
22     for(i = 0; i < n; i++) {
23         tat[i] = wt[i] + bt[i];
24     }
25     avg_wt += wt[i];
```

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```
Enter number of processes: 3
Enter Burst Time for Process 1: 5
Enter Burst Time for Process 2: 3
Enter Burst Time for Process 3: 8
```

Process	Burst Time	Waiting Time	Turnaround Time
P1	5	0	5
P2	3	5	8
P3	8	8	16

```
Average Waiting Time = 4.33
Average Turnaround Time = 9.67
```

```
-----
Process exited after 34.78 seconds with return value 0
Press any key to continue . . .
```

```

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Enter number of processes: 4
Enter Burst Time for Process 1: 6
Enter Burst Time for Process 2: 8
Enter Burst Time for Process 3: 7
Enter Burst Time for Process 4: 3

Process Burst Time      Waiting Time      Turnaround Time
P4       3              0              3
P1       6              3              9
P3       7              9             16
P2       8             16             24

Average Waiting Time = 7.00
Average Turnaround Time = 13.00
-----
Process exited after 6.853 seconds with return value 0
Press any key to continue . . .

```

```
C:\Users\91769\OneDrive\Desktop\study_files\OS\C lab\EXP-5.cpp - [Executing] - Dev-C++ 5.11
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(globals)
Project Classes Debug
EXP-7.cpp EXP-8.cpp
EXP-13.cpp EXP-12.cpp EXP-14.cpp EXP-15.cpp EXP-16.cpp
1 #include <stdio.h>
2
3 int main() {
4     int p[20], bt[20], pr[20], wt[20], tat[20];
5     int n, i, j, temp;
6     float avg_wt = 0, avg_tat = 0;
7
8     printf("Enter number of processes: ");
9     scanf("%d", &n);
10
11     for(i = 0; i < n; i++) {
12         p[i] = i + 1;
13
14         printf("Enter Burst Time for Process %d: ", p[i]);
15         scanf("%d", &bt[i]);
16
17         printf("Enter Priority for Process %d: ", p[i]);
18         scanf("%d", &pr[i]);
19     }
20
21     // Sorting according to priority
22     for(i = 0; i < n; i++) {
23         for(j = i + 1; j < n; j++) {
24             if(pr[i] > pr[j]) {
25                 // Swap p[i] and p[j]
26                 temp = p[i];
27                 p[i] = p[j];
28                 p[j] = temp;
29                 // Swap bt[i] and bt[j]
30                 temp = bt[i];
31                 bt[i] = bt[j];
32                 bt[j] = temp;
33                 // Swap pr[i] and pr[j]
34                 temp = pr[i];
35                 pr[i] = pr[j];
36                 pr[j] = temp;
37             }
38         }
39     }
40
41     // Calculating waiting and turnaround times
42     wt[0] = 0;
43     for(i = 1; i < n; i++) {
44         wt[i] = wt[i-1] + bt[i-1];
45     }
46
47     for(i = 0; i < n; i++) {
48         tat[i] = wt[i] + bt[i];
49         avg_wt += wt[i];
50         avg_tat += tat[i];
51     }
52
53     avg_wt /= n;
54     avg_tat /= n;
55
56     printf("Average Waiting Time = %.2f\n", avg_wt);
57     printf("Average Turnaround Time = %.2f\n", avg_tat);
58
59     printf("Process exited after 40.86 seconds with return value 0\n");
60     printf("Press any key to continue . . .\n");
61     getch();
62 }
```

```
C:\Users\91769\OneDrive\Desktop\study_files\OS\C lab\EXP-5.cpp - [Executing] - Dev-C++ 5.11
Enter number of processes: 4
Enter Burst Time for Process 1: 10
Enter Priority for Process 1: 3
Enter Burst Time for Process 2: 5
Enter Priority for Process 2: 1
Enter Burst Time for Process 3: 8
Enter Priority for Process 3: 4
Enter Burst Time for Process 4: 6
Enter Priority for Process 4: 2

Process Priority      Burst Time      Waiting Time      Turnaround Time
P2      1      5      0      5
P4      2      6      5      11
P1      3      10     11     21
P3      4      8     21     29

Average Waiting Time = 9.25
Average Turnaround Time = 16.50
-----
Process exited after 40.86 seconds with return value 0
Press any key to continue . . .
```


C:\Users\91769\OneDrive\Desktop\study_files\OS\C lab\EXP-6.cpp - [Executing] - Dev-C++ 5.11

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(globals)

Project Classes Debug

EXP-7.cpp EXP-8.cpp EXP-9.cpp EXP-10.cpp EXP-11.cpp
EXP-13.cpp EXP-12.cpp EXP-14.cpp EXP-15.cpp EXP-16.cpp EXP-17.cpp EXP-18.cpp EXP-19.cpp EXP-20.cpp EXP-1.cpp EXP-2.cpp EXP-3.cpp EXP-4.cpp EXP-5.cpp EXP-6.cpp

```
1 #include <stdio.h>
2
3 int main() {
4     int pid[10], bt[10], pr[10], wt[10], tat[10], rt[10];
5     int n, i, time = 0, completed = 0;
6     int highest, temp;
7     float avg_wt = 0, avg_tat = 0;
8
9     printf("Enter number of processes: ");
10    scanf("%d", &n);
11
12    for(i = 0; i < n; i++) {
13        pid[i] = i + 1;
14
15        printf("Enter Burst Time for Process %d: ", pid[i]);
16        scanf("%d", &bt[i]);
17
18        printf("Enter Priority for Process %d: ", pid[i]);
19        scanf("%d", &pr[i]);
20
21        rt[i] = bt[i];
22    }
23
24    pr[9] = 9999;
```

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Enter Priority for Process 1: 2
Enter Burst Time for Process 2: 3
Enter Priority for Process 2: 1
Enter Burst Time for Process 3: 5
Enter Priority for Process 3: 3

Process	Burst Time	Priority	Waiting Time	Turnaround Time
P1	4	2	3	7
P2	3	1	0	3
P3	5	3	7	12

Average Waiting Time = 3.33
Average Turnaround Time = 7.33

Process exited after 34.87 seconds with return value 0
Press any key to continue . . .

C:\Users\91769\OneDrive\Desktop\study_files\OS\C lab\EXP-7.cpp - [Executing] - Dev-C++ 5.11

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TDM-GCC 4.9.2 64-bit Release

(globals)

Project Classes Debug

EXP-13.cpp

EXP-12.cpp

EXP-14.cpp

EXP-15.cpp

EXP-16.cpp

EXP-17.cpp

EXP-18.cpp

EXP-19.cpp

EXP-20.cpp

EXP-1.cpp

EXP-2.cpp

EXP-3.cpp

EXP-4.cpp

EXP-5.cpp

EXP-7.cpp

EXP-8.cpp

EXP-9.cpp

EXP-10.cpp

EXP-11.cpp

```
1  #include <stdio.h>
2
3  int main() {
4      int bt[20], p[20], wt[20], tat[20];
5      int n, i, j, temp;
6      float avg_wt = 0, avg_tat = 0;
7
8      printf("Enter number of processes: ");
9      scanf("%d", &n);
10
11     for(i = 0; i < n; i++) {
12         p[i] = i + 1;
13
14         printf("Enter Burst Time for Process %d: ", p[i]);
15         scanf("%d", &bt[i]);
16     }
17
18     // Sorting burst times in ascending order
19     for(i = 0; i < n; i++) {
20         for(j = i + 1; j < n; j++) {
21
22             if(bt[i] > bt[j]) {
23
24                 temp = bt[i];
25                 bt[i] = bt[j];
```

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```
Enter number of processes: 4
Enter Burst Time for Process 1: 6
Enter Burst Time for Process 2: 2
Enter Burst Time for Process 3: 8
Enter Burst Time for Process 4: 3
```

Process	Burst Time	Waiting Time	Turnaround Time
P2	2	0	2
P4	3	2	5
P1	6	5	11
P3	8	11	19

```
Average Waiting Time = 4.50
Average Turnaround Time = 9.25
```

```
-----
Process exited after 16.7 seconds with return value 0
Press any key to continue . . .
```

C:\Users\91769\OneDrive\Desktop\study_files\OS\C lab\EXP-8.cpp - [Executing] - Dev-C++ 5.11

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TDM-GCC 4.9.2 64-bit Release

(globals)

Project Classes Debug EXP-13.cpp EXP-12.cpp EXP-14.cpp EXP-15.cpp EXP-16.cpp EXP-17.cpp EXP-18.cpp EXP-19.cpp EXP-20.cpp EXP-1.cpp EXP-2.cpp EXP-3.cpp EXP-4.cpp EXP-5.cpp EXP-6.cpp EXP-7.cpp EXP-8.cpp EXP-9.cpp EXP-10.cpp EXP-11.cpp

```
1 #include <stdio.h>
2
3 int main() {
4     int i, n, tq;
5     int bt[20], rt[20];
6     int wt[20], tat[20];
7     int time = 0, remain;
8     float avg_wt = 0, avg_tat = 0;
9
10    printf("Enter number of processes: ");
11    scanf("%d", &n);
12
13    remain = n;
14
15    for(i = 0; i < n; i++) {
16        printf("Enter Burst Time for Process %d: ", i + 1);
17        scanf("%d", &bt[i]);
18
19        rt[i] = bt[i];
20    }
21
22    printf("Enter Time Quantum: ");
23    scanf("%d", &tq);
24
25    i = 0;
```

Enter number of processes: 3
Enter Burst Time for Process 1: 10
Enter Burst Time for Process 2: 5
Enter Burst Time for Process 3: 8
Enter Time Quantum: 2

Process	Burst Time	Waiting Time	Turnaround Time
P1	10	13	23
P2	5	10	15
P3	8	13	21

Average Waiting Time = 12.00
Average Turnaround Time = 19.67

Process exited after 7.932 seconds with return value 0
Press any key to continue . . .

